

Course Title: Cryptography and Secure Communication

	2.5. Implementing HMAC for Message Authentication Chapter - 3. Secure Web & Email Communication 3.1. SSL/TLS and HTTPS Protocol 3.2. Configuring SSL Certificates for Web Servers 3.3. Email Encryption Standards (PGP, S/MIME) 3.4. Implementing End-to-End Email Encryption 3.5. Analyzing SSL/TLS Traffic Using Wireshark Chapter - 4. Cryptographic Attacks & Countermeasures 4.1. Brute-Force and Dictionary Attacks on Encryption 4.2. Man-in-the-Middle (MITM) Attacks and Prevention 4.3. Rainbow Table and Collision Attacks 4.4. Side-Channel Attacks on Cryptographic Systems 4.5. Implementing Secure Key Exchange Protocols Suggested Practical Exercises: - Encrypting and Decrypting Messages Using AES & RSA - Generating and Verifying Digital Signatures - Setting Up SSL Certificates for Secure Communication - Implementing Secure Email Communication Using PGP - Conducting Cryptographic Attacks and Mitigation
Reference Books	1. Cryptography and Network Security: Principles and Practice – <i>William Stallings</i> 2. Applied Cryptography: Protocols, Algorithms, and Source Code in C – <i>Bruce Schneier</i> 3. Introduction to Modern Cryptography – <i>Jonathan Katz & Yehuda Lindell</i> 4. Understanding Cryptography: A Textbook for Students and Practitioners – <i>Christof Paar & Jan Pelzl</i> 5. Practical Cryptography in Python – <i>Seth Nielson & Christopher Monson</i> 6. Network Security and Cryptography – <i>Bernard Menezes</i> 7. Handbook of Applied Cryptography – <i>Alfred J. Menezes, Paul C. van Oorschot, Scott A. Vanstone</i> 8. Modern Cryptography: Theory and Practice – <i>Wenbo Mao</i> 9. Beginning Cryptography with Java – <i>David Hook</i> 10. SSL and TLS: Designing and Building Secure Systems – <i>Eric Rescorla</i>
Teaching Methodology	Class Work, Discussion, Self-Study, Case study, Seminars and/or Assignments
Evaluation Method	50% Internal assessment. 50% External assessment.